Cuda invalid configuration error when using the statevector GPU simulato

https://github.com/Qiskit/qiskit-aer/issues/1646

ROW 10

Cuda invalid configuration error when using the statevector GPU simulator with a large number of qubits #1646

New issue

Closed

HaoTy opened on Nov 6, 2022 · edited by HaoTy

Edits ▾ ···

Informations

- **Qiskit Aer version:**

```
qiskit                    0.39.2
qiskit-aer                0.11.1
qiskit-aer-gpu            0.11.1
qiskit-ibmq-provider      0.19.2
qiskit-optimization       0.4.0
qiskit-terra              0.22.2
```

- **Python version:** 3.9.12
- **Operating system:** CentOS Stream 8
- **GPU:** A100-SXM4-80GB
- **CUDA version:** 11.7.0

What is the current behavior?

Cuda gives an invalid configuration error when using the statevector GPU simulator with a large number of qubits (more than 30 in my case).

```
qiskit.exceptions.QiskitError: 'Circuit execution failed: ERROR:  [Experiment 0] for_each: failed to syn
```

Note that `AerPauliExpectation` is used and not until n=36 the memory becomes insufficient:

```
qiskit.exceptions.QiskitError: 'Circuit execution failed: ERROR:  [Experiment 0] Insufficient memory to
```

Steps to reproduce the problem

```
import numpy as np
import networkx as nx
from qiskit_aer import AerSimulator
from qiskit.algorithms import QAOA
from qiskit.algorithms.optimizers import SPSA
from qiskit.utils import QuantumInstance
from qiskit.opflow import AerPauliExpectation
from qiskit_optimization.applications import Maxcut
from qiskit.utils import algorithm_globals

n = 32
p = 1
seed = 0

graph = nx.random_regular_graph(3, n, seed)
problem = Maxcut(graph).to_quadratic_program()
H, offset = problem.to_ising()

algorithm_globals.random_seed = seed
backend = AerSimulator(method="statevector", device="GPU")
quantum_instance = QuantumInstance(
    backend=backend,
    seed_simulator=seed,
    seed_transpiler=seed,
)
```

Assignees

doichanj

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

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Participants

```
algorithm_globals.random_seed = seed
backend = AerSimulator(method="statevector", device="GPU")
quantum_instance = QuantumInstance(
    backend=backend,
    seed_simulator=seed,
    seed_transpiler=seed,
)

algorithm = QAOA(
    SPSA(),
    reps=p,
    quantum_instance=quantum_instance,
    expectation=AerPauliExpectation(),
)
algorithm._check_operator_ansatz(H)
energy_evaluation= algorithm.get_energy_evaluation(H)
parameters = 2 * np.pi * np.random.default_rng(seed).random(2 * p)
energy_evaluation(parameters)
```

What is the expected behavior?

The simulator should be able to evaluate the QAOA energy like it does when n=30 or below.

HaoTy added bug on Nov 6, 2022

hhorii assigned doichanj on Nov 7, 2022

doichanj on Nov 7, 2022 Collaborator

This issue is caused because Aer tries allocating chunks in all free memory on GPU for hybrid parallelization. I will make fix to allocate chunks about 80% of free memory.

doichanj mentioned this on Nov 7, 2022

Fix for GPU simulator, memory allocation #1650

hhorii closed this as completed on Nov 15, 2022